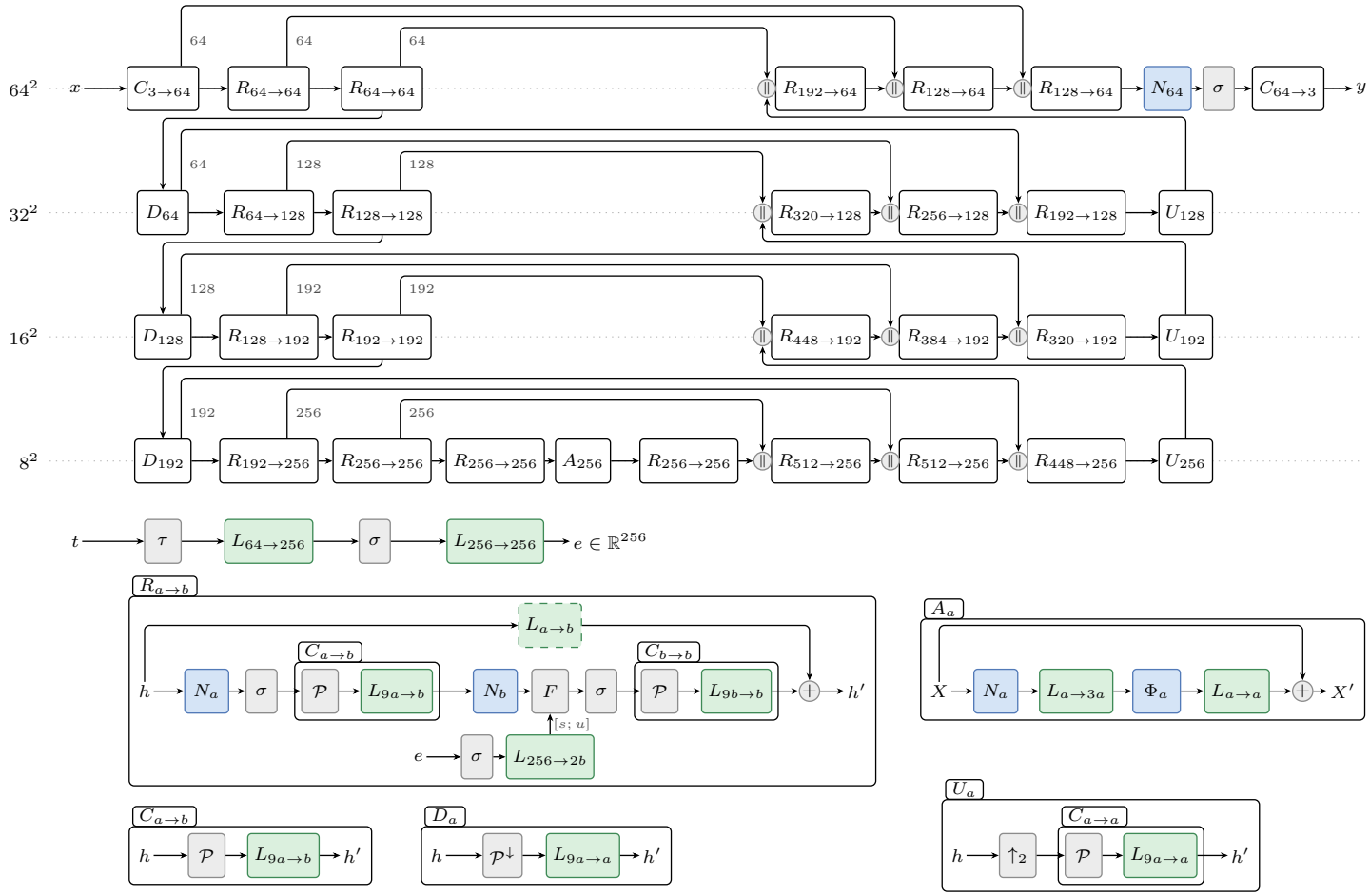
U-Net, CelebA 64×64 $s_\theta : \mathbb{R}^{3\times64\times64} \times \mathbb{R} \to \mathbb{R}^{3\times64\times64}$, $(x, t) \mapsto y$, 20 972 867 parameters.



□ object of a sparse-transformers class (S_κ, $\kappa \le n$) □ sparsifiable, not trivially mappable: none □ parameter-free
□ substitutable (attention-free attention, dynamic tanh) □ identity for $a = b$

	domain → codomain	definition	parameters	sparse-transformers
σ	$\mathbb{R} \to \mathbb{R}$, elementwise	$x \mapsto x/(1+e^{-x})$	0	□
τ	$\mathbb{R} \to \mathbb{R}^{64}$	$t \mapsto (\cos\omega_i t)_{i<32} \oplus (\sin\omega_i t)_{i<32}$, $\omega_i = 10^{-i/8}$	0	□
$L_{n\to m}$	$\mathbb{R}^n \to \mathbb{R}^m$ per position	$x \mapsto Wx + w_0$, $W \in \mathbb{R}^{m\times n}$, $w_0 \in \mathbb{R}^m$	$nm + m$	□ `SparseLinear`, $\kappa = n$
$\mathcal{P}$	$\mathbb{R}^{a\times h\times w} \to \mathbb{R}^{9a\times h\times w}$	$(\mathcal{P}h)_{(c,p,q),i,j} = h_{c,i+p,j+q}$, $p, q \in \{-1, 0, 1\}$, zero padding	0	□
$\mathcal{P}^\downarrow$	$\mathbb{R}^{a\times h\times w} \to \mathbb{R}^{9a\times h/2\times w/2}$	$(\mathcal{P}^\downarrow h)_{(c,p,q),i,j} = h_{c,2i+p,2j+q}$, p, q and padding as for $\mathcal{P}$	0	□
$\uparrow_2$	$\mathbb{R}^{a\times h\times w} \to \mathbb{R}^{a\times 2h\times 2w}$	$(\uparrow_2 h)_{c,i,j} = h_{c,\lfloor i/2\rfloor,\lfloor j/2\rfloor}$	0	□
N_a	$\mathbb{R}^{a\times h\times w} \to \mathbb{R}^{a\times h\times w}$	$\gamma_c\,(h_{c,i,j} - \mu_g)/\sqrt{v_g + 10^{-5}} + \beta_c$; μ_g, v_g: mean, variance of group $g \ni c$ ($a/32$ channels, all positions)	$2a$	□ `DyT`
F	$\mathbb{R}^{b\times h\times w} \times \mathbb{R}^{2b} \to \mathbb{R}^{b\times h\times w}$	$(z, [s; u]) \mapsto z \odot (1 + s) + u$, s, u broadcast over positions	0	□
Φ_a	$\mathbb{R}^{3a\times h\times w} \to \mathbb{R}^{a\times h\times w}$	$[Q_1; K_1; V_1; \ldots; Q_4; K_4; V_4] \mapsto [H_1; \ldots; H_4]$, $Q_j, K_j, V_j, H_j \in \mathbb{R}^{a/4\times hw}$; $H_j = V_j\,\mathrm{softmax}(K_j^\top Q_j/\sqrt{a/4})$, softmax over keys	0	□ `AFTConv`, `AFTRecurrence` (causal), for $L_{a\to a}\,\Phi_a\,L_{a\to3a}$
$\|$	$\mathbb{R}^{a\times h\times w} \times \mathbb{R}^{a'\times h\times w} \to \mathbb{R}^{(a+a')\times h\times w}$	$(h, h_{\text{skip}}) \mapsto [h; h_{\text{skip}}]$, h from the lane, h_{skip} from the arc	0	□
$C_{a\to b}$	$\mathbb{R}^{a\times h\times w} \to \mathbb{R}^{b\times h\times w}$	$L_{9a\to b} \circ \mathcal{P}$	$9ab + b$	
D_a	$\mathbb{R}^{a\times h\times w} \to \mathbb{R}^{a\times h/2\times w/2}$	$L_{9a\to a} \circ \mathcal{P}^\downarrow$	$9a^2 + a$	
U_a	$\mathbb{R}^{a\times h\times w} \to \mathbb{R}^{a\times 2h\times 2w}$	$C_{a\to a} \circ \uparrow_2$	$9a^2 + a$	
$R_{a\to b}$	$\mathbb{R}^{a\times h\times w} \times \mathbb{R}^{256} \to \mathbb{R}^{b\times h\times w}$	$(h, e) \mapsto S(h) + C_{b\to b}\,\sigma\,F\left(N_b\,C_{a\to b}\,\sigma\,N_a h,\ L_{256\to2b}\,\sigma(e)\right)$; $S = L_{a\to b}$ if $a \ne b$, else $S = \mathrm{id}$	$2a + 9ab + 9b^2 + 518b + [a \ne b](ab + b)$	
A_a	$\mathbb{R}^{a\times h\times w} \to \mathbb{R}^{a\times h\times w}$	$X \mapsto X + L_{a\to a}\,\Phi_a\,L_{a\to3a}\,N_a X$	$4a^2 + 6a$	

stage	blocks	parameters
time	$L_{256\to256}\,\sigma\,L_{64\to256}\,\tau$	82 432
stem 64^2	$C_{3\to64}$	1 792
encoder 64^2	$R_{64\to64}$, $R_{64\to64}$	214 016
$64^2 \to 32^2$	D_{64}	36 928
encoder 32^2	$R_{64\to128}$, $R_{128\to128}$	657 408
$32^2 \to 16^2$	D_{128}	147 584
encoder 16^2	$R_{128\to192}$, $R_{192\to192}$	1 440 832
$16^2 \to 8^2$	D_{192}	331 968
encoder 8^2	$R_{192\to256}$, $R_{256\to256}$	2 527 360
middle 8^2	$R_{256\to256}$, A_{256}, $R_{256\to256}$	2 889 216

stage	blocks	parameters
decoder 8^2	$R_{512\to256}$, $R_{512\to256}$, $R_{448\to256}$	5 939 328
$8^2 \to 16^2$	U_{256}	590 080
decoder 16^2	$R_{448\to192}$, $R_{384\to192}$, $R_{320\to192}$	3 508 416
$16^2 \to 32^2$	U_{192}	331 968
decoder 32^2	$R_{320\to128}$, $R_{256\to128}$, $R_{192\to128}$	1 626 240
$32^2 \to 64^2$	U_{128}	147 584
decoder 64^2	$R_{192\to64}$, $R_{128\to64}$, $R_{128\to64}$	497 856
head 64^2	$C_{64\to3}\,\sigma\,N_{64}$	1 859
total		20 972 867

FAN-IN.

$$S_\kappa : \mathbb{R}^n \to \mathbb{R}^m,\quad S_\kappa(x) = (M \odot W)x + w_0,\quad M \in \{0,1\}^{m\times n},\quad \{j : M_{ij} = 1\} \sim \mathrm{Unif}\tbinom{[n]}{\kappa}\ \ \forall i,\quad 1 \le \kappa \le n,\quad S_n = L_{n\to m};$$

$$G_\kappa = (V, E) :\quad (j \to i) \in E \iff M_{ij} = 1,\quad k_i^- = \sum_j M_{ij} = \kappa,\quad k_j^+ = \sum_i M_{ij},\quad \mathbb{E}\,k_j^+ = \tfrac{m\kappa}{n},\quad k_v = k_v^- + k_v^+;$$

$$G = \bigcup\nolimits_l G_{\kappa_l} \text{ over the affine maps } L_l = S_{\kappa_l} \text{ of } s_\theta\ (\kappa_l = n_l),\ k_v^\pm \text{ summed over the maps at } v :\quad K_G := \max_v k_v \ge \max_v k_v^- = \max_l \kappa_l.$$